Food chains

BACKGROUND

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

In an ecosystem, one organism depends on another to survive. Producers make food, which is then eaten by consumers, which is then broken down by decomposers. Consumers and decomposers turn this food into the energy they need to survive. This creates what scientists call a **food chain**.

A food chain can be represented using pictures or words, but it always needs arrows. The arrows show what is eaten by what and the **flow of energy** along the food chain. They point away from the thing that is being eaten. For example:

plant → grasshopper → snake → eagle

Sometimes food chains show decomposers. Sometimes they show just producers and consumers, or just consumers.

☞ **1** Read these statements and then write a food chain for each. Two examples have been done for you. Hint: Make sure your organisms are in the correct order. The correct order may not be the order in which they appear in the statement. Check your answers by replacing each arrow in the food chain with 'is eaten by'.

a) A scorpion is eaten by a meerkat, which is eaten by a cobra, which is eaten by a vulture.

scorpion → meerkat → cobra → vulture

b) A tiger eats a deer, which eats a plant.

plant → deer → tiger

c) A ferret eats a rabbit, which eats grass.

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d) A brown bear eats a salmon, which eats plankton.

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Food chains CONT'D

e) Seaweed is eaten by a small fish, which is eaten by a tuna, which is eaten by a cormorant.

f) A lynx eats a raccoon, which eats a rat, which eats an insect.

g) A wolf eats a woodpecker, which eats a frog, which eats a fly.

h) Grain is eaten by a fieldmouse, which is eaten by a weasel, which is eaten by a cougar.

i) A shark eats a jewfish, which eats a whiting, which eats a prawn, which eats a worm.

j) Fruit is eaten by a grub, which is eaten by a monkey, which is eaten by a hyena.

k) A crab eats a sea snail, which eats a starfish, which eats a mussel, which eats plankton.

Food chains are not the best way to describe the relationships in most ecosystems. They are too simple. Ecosystems usually involve many species that are connected in all sorts of ways.

For example, this food chain is from an African savannah: grass → zebra → lion. There are many savannah animals that consume grass, and many that are eaten by lions. Birds also eat the remains of zebras, and decomposers like bacteria, flies and microbes eat them as well. The best way to show all of these relationships is using a **food web**.

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